3.4 · Weigh Your Agent

Week 2 · Wednesday Jul 22 — Measurement, Evaluation, and the Startup Pitch

Generative AI & Sustainability · Munich Summer School 2026

Session goal

Replace one *cited* number in your agent's sustainability label with a number you **measured yourself**.

By the end, you should have:

- two local models running on one laptop per team
- one representative request from your agent map, run ~20 times per model
- one `emissions.csv` with energy and CO_2e per model
- one measured row in your evaluation plan and scorecard

The task

Pick **one representative request** from your Monday agent map — the single interaction your prototype demonstrates (e.g., "*find me a quiet room with outlets*").

Run it through two models and weigh each run:

Tier	Model	Size on disk
Small	`qwen2.5:0.5b`	~0.4 GB
Mid	`gemma2:2b`	~1.6 GB

- Repeat the request **~20 times per model**, varying the wording slightly, so the measurement isn't noise.

Two numbers, two jobs

Energy

CodeCarbon watches the machine while your loop runs and writes one row per model:
energy (kWh), emissions (kg CO_2e), duration.

Quality

Correct? useful? tone? — scored on 5 answers per model by hand.

- **Energy without quality is half a measurement. A model that is cheap and wrong costs the most.**

Setup

Five short steps, mostly one command each. Do this before 1:15 if you can.

What you are installing and why

Piece	What it is	Why you need it
Python	the language the 15-line script is written in	runs the loop
Ollama	an app that runs LLMs **on your laptop**	no API, no cloud, so the energy is *yours* to measure
codecarbon	a Python package	measures energy and converts to CO_2e

- Everything runs locally. No account, no API key, no data leaves the laptop.

Step 1 · Install Python

macOS

Download the installer from python.org/downloads and open it.

Click through with the defaults.

Windows

Download the installer from python.org/downloads and open it.

☑ **Check "Add python.exe to PATH"** on the first screen before clicking Install.

- **Windows: if you miss that checkbox, nothing below will work. Re-run the installer and check it.**

Step 2 · Install the Python packages

Open a terminal first — Terminal on macOS (Cmd + Space), PowerShell on Windows — then run one line:

macOS

```
python3 -m pip install codecarbon ollama
```

Windows

```
python -m pip install codecarbon ollama
```

Package	What it does
`codecarbon`	measures energy, converts to CO_2e
`ollama`	lets your script talk to the Ollama app

- From here on: macOS uses `python3`, Windows uses `python`. Everything else is identical.

Step 3 · Install the Ollama app

The `ollama` package you just installed is only the **client**. The thing that actually runs models is a small desktop app.

Download it from ollama.com/download, open it, and leave it running.

macOS

Drag **Ollama** to Applications and launch it once.

Windows

Run `OllamaSetup.exe`; it starts automatically after installing.

- A small icon in the menu bar / system tray means the model server is running.

Step 4 · Check that both halves are there

macOS

```
python3 --version
ollama --version
```

Windows

```
python --version
ollama --version
```

- Both commands should print a version number.
- If one says *command not found*, close the terminal, reopen it, and try again — an installer's PATH change only reaches **new** terminals.

Step 5 · Pull the models

Same command on both systems:

```
ollama pull qwen2.5:0.5b
ollama pull gemma2:2b
```

- The two pulls download ~2 GB total. **Start them now** (preferably on eduroam wifi)
- `ollama list` shows what finished downloading.

Run the measurement

One folder, one file, one command.

Make a folder, get the script

Run each line separately:

```
cd Documents
mkdir weigh-your-agent
cd weigh-your-agent
```

You do not have to type the script. It is in the course repo:

```
public/scripts/weigh.py
```

Open it on GitHub, download it into `weigh-your-agent`, and edit the `PROMPTS` list.

- Your terminal must be **in that folder** when you run the script.
- Same project-boundary habit as yesterday: script, CSV, and results stay together.

Reading the script (no Python needed)

- The list at the top is the only part you must edit: ~20 wordings of your one request.
- The outer loop repeats everything once per model.
- The warm-up call loads the model into memory *before* the tracker starts — otherwise you measure the loading, not the answering.
- The `with EmissionsTracker(...)` block is the stopwatch: everything inside it gets weighed.
- `eval_count` = tokens generated · `eval_duration` = nanoseconds spent generating.

Run it

macOS

```
python3 weigh.py
```

Windows

```
python weigh.py
```

- Expect a few minutes per model. Leave the laptop plugged in and don't switch tasks — you are measuring this machine.
- When it finishes, `emissions.csv` is in the folder: **one row per model**, with `energy_consumed` (kWh), `emissions` (kg CO_2e), `duration`.

Scale the numbers two ways

```
kWh per 1,000 requests = energy_consumed / len(PROMPTS) × 1000

kWh per 1,000 tokens   = energy_consumed / gen_tokens × 1000
```

- Per request is what a user costs you.
- Per token is fairer across models — the bigger model isn't penalized just for writing longer answers.
- tok/s is throughput: average the printed values across the loop.
- **The small-vs-large gap in tok/s and kWh is half your routing argument.**

Judge quality, not just cost

While one teammate runs the loop, the others score 5 answers from each model with yesterday's rubric: correct? useful? tone?

Then answer the only question that matters:

- **Does the small model pass *your* quality bar? If yes — what justifies routing anything to the bigger tiers?**

Deliverable — one measured row, cited honestly

Goes into today's evaluation plan **and** the sustainability scorecard.

Task	Model	tok/s	kWh / 1,000 req	g CO_2e / 1,000 req	kWh / 1,000 tok	Passes quality bar?
quiet-room request	qwen2.5:0.5b	*measured*	*measured*	*measured*	*measured*	yes / no
quiet-room request	gemma2:2b	*measured*	*measured*	*measured*	*measured*	yes / no

Footnote for your pitch slide:

> *"Measured with CodeCarbon on [laptop model], July 22, 2026, n=20; German grid intensity."*

Honesty notes — these are features, not bugs

- On many laptops CodeCarbon **estimates** CPU/GPU power from hardware specs rather than reading meters. Say so in the footnote.
    - Week 1's lesson: methodology spreads of 14× exist *between published papers*. Your transparent estimate still beats a copied headline.
- A laptop is not a datacenter: no PUE, no cooling overhead, different hardware. Your number is a **lower bound and a ratio** — the *ratio* is the robust finding, and the ratio is what justifies your routing tier.
- This morning's paper measured whole **agent loops**, not single calls. If time remains, run your prompt through a 2–3 step loop (retrieve → answer → summarize) and watch the multiplier appear in your own CSV.

Fallbacks

- `gemma2:2b` **still downloading:** start the loop with `qwen2.5:0.5b` alone and add the second row when the pull finishes.
- **Ollama won't install:** use the shared Colab notebook (GPU runtime; CodeCarbon works there) — link on the course pad.
- **Python won't install:** measure on a teammate's machine; one measurement per team is the requirement.
- **Everything fails:** the instructor's machine runs one team-supplied prompt set live; you still record and cite the measurement.

Common troubleshooting

Symptom	Fix
`python: command not found`	reopen the terminal; on Windows re-run the installer with **Add to PATH**
`No module named codecarbon`	you installed with a different Python — rerun `python3 -m pip install codecarbon ollama`
`connection refused`	the Ollama app isn't running — launch it from Applications / Start menu
`model not found`	`ollama list`, then re-run the missing `ollama pull`
laptop crawls on `gemma2:2b`	that *is* a result — note the tok/s and keep going

References

- Ollama download · model library
- Python downloads
- CodeCarbon documentation
- Ollama Python library

Setup details checked **22 July 2026**. Installers and commands change; use the linked official documentation if a screen differs.